

# Complex Analysis – 3

Принцип аргумента, теорема Руше и подсчёт нулей

Домашнее задание

1 сентября 2026 года

**Теоретический минимум.** The winding number and the argument principle for meromorphic functions; Rouché’s theorem in the classical and in the symmetric form  $|f - g| < |f| + |g|$ ; Hurwitz’s theorem; the open mapping theorem and the maximum principle as its corollary; Jensen’s formula  $\frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{i\theta})| d\theta = \log |f(0)| + \sum \log \frac{r}{|a_k|}$  as the quantitative form of zero counting.

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1. Count the zeros of  $z^5 + 3z^3 + 7$  in the disc  $|z| < 2$ , and determine how many of them lie in the right half-plane.
2. Prove the symmetric form of Rouché’s theorem: if  $|f(z) - g(z)| < |f(z)| + |g(z)|$  on  $\gamma$ , then  $f$  and  $g$  have the same number of zeros inside. Show that it is strictly stronger than the classical form.
3. Prove the open mapping theorem from the argument principle, and deduce both the maximum modulus principle and the fact that  $|f|$  has no interior local minimum unless  $f$  vanishes there.
4. (S.T. Yau CSMC 2019)
  - a) Show that there is no holomorphic function  $f$  on  $\mathbb{C} \setminus \{1, -1\}$  with  $f'(z) = \frac{1}{(z^2-1)^{2019}}$  for all  $z$ .
  - b) Show that there are a set  $L \subset \mathbb{C}$  of Hausdorff dimension 1 and a holomorphic function  $F$  on  $\mathbb{C} \setminus L$  with  $F'(z) = \frac{1}{(z^2-1)^{2019}}$ .
5. Prove Jensen’s formula and use it to bound the number of zeros of a bounded holomorphic function in  $|z| < r$ ; deduce the Blaschke condition  $\sum(1 - |a_n|) < \infty$ .

6. Let  $s_n(z) = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!}$ . Prove that for every  $R > 0$  the polynomial  $s_n$  has no zeros in  $|z| \leq R$  once  $n$  is large enough, and explain what Hurwitz's theorem has to do with it.
7. (S.T. Yau CSMC 2023) Suppose  $f$  is holomorphic in a neighbourhood  $\Omega$  of the closed unit disc except for a pole at a point  $z_0$  on the unit circle. Show that if  $\sum_{n \geq 0} a_n z^n$  is the power series of  $f$  in  $\mathbb{D}$ , then  $\lim_{n \rightarrow \infty} \frac{a_n}{a_{n+1}} = z_0$ .
8. Prove that for a polynomial  $p$  of degree  $n$  the number of zeros in a half-plane can be read off the variation of  $\arg p$  along the boundary line; use this to determine how many roots of  $z^4 + z^3 + 4z^2 + 2z + 3$  have positive real part.
9. Prove the Gauss–Lucas theorem by the argument principle, and show that if all zeros of  $p$  lie in  $|z| \leq 1$  then all zeros of  $p'$  do too, with equality of the extremal cases exactly for  $p(z) = c(z - a)^n$ .
10. Let  $f$  be holomorphic on a neighbourhood of  $\overline{\mathbb{D}}$  and injective on the unit circle. Prove that  $f$  is injective on  $\mathbb{D}$  and that  $f(\mathbb{D})$  is the interior of the Jordan curve  $f(\partial\mathbb{D})$ .